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Food waste in hospitality and food services: A systematic literature review and framework development approach

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ABSTRACT

This study critically analyzes the state-of-the-art of food waste in the hospitality and food services (HaFS) sector body of literature. It uses a systematic literature review (SLR) approach implemented through search, evaluation, and synthesis of peer-reviewed articles. The primary focus is on uncovering key research themes and gaps in the extant knowledge to edify and advance a future research agenda. Content analysis is used to aggregate the selected articles around nine themes representing various aspects of food waste. The themes range from the causes of waste generation to leftover handling and waste reduction. Additionally, extensive research profiling is undertaken to present summary statistics of the selected articles in terms of research design, methods of data analysis, variables investigated, and the theoretical lens used. The SLR raises some interesting research questions and offers actionable inferences for practice. The study concludes with a framework that brings the findings together to inform future empirical research in the area.
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1. Introduction

Food waste is defined as the use of food meant for consumption by humans for non-consumption purposes, the redirection of food to feed animals, or the disposal of edible food (FAO, 2014). It includes the edible as well as inedible parts of food that get removed from the food supply chain and which can be recovered or managed through disposal (Östergren et al., 2014). Furthermore, food waste can be grouped into three different types: (a) avoidable waste, which is food that was edible at some point in time but has become inedible by the time it reaches disposal; (b) unavoidable waste, which refers to certain items, like eggshells, that are not edible; and (c) potentially avoidable food waste, which applies to particular wastes that are consumed at times, but not always, such as potato skins (Papargyropoulou et al., 2014). Scholars also argue that these three broad classifications of food waste may differ in different cultural contexts (Liu, 2014).

In the prior literature, some scholars have used the term "food loss" as a synonym for food waste (e.g., Betz et al., 2015). However, others have distinguished between the two, where "food loss" represents food waste at the start of the value-add chain, while "food waste" refers to food lost at the end of the chain (Parfitt et al., 2010). Taking both views into consideration, we treated food loss and food waste as the same. We argue that food loss or waste is observed at different stages of the food chain, including the consumption stage (Martin-Rios et al., 2018). Waste at the stage of consumption can be generated at the household level (home dining) or at the hospitality level (out-of-home dining). The hospitality sector, representing out-of-home dining, can be further subdivided into three parts: non-commercial, commercial, and other food services (Betz et al., 2015).

Hospitality sector food waste is fast becoming a key concern,

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given that its contribution to total food waste has been nearly 12% in the recent past (Tostivint et al., 2016). Furthermore, with the increasing trend of out-of-home dining, spurred by growth in incomes and tourism, hospitality waste has become a significant issue for both developed and developing countries (Wang et al., 2017). Although the amount of food-related waste generated in this sector is frequently discussed in the media, it has not yet got sufficient academic attention (Filimonau and De Coteau, 2019). Other scholars have also acknowledged that the issue of food-related waste in this sector has been investigated less rigorously despite being recognized as a key challenge (Principato et al., 2018; Wang et al., 2017).

The extant literature examining food waste in the hospitality sector has focused on a diversity of topics, such as food waste quantification, the composition of waste; waste handling; doggy bags; the attitudes of consumers; demographic factors; governmental regulations, interventions, and nudges; composting; and landfills. While scholars have ventured into the visible spectrum of the sub-domains of food waste, most of their findings remain in different silos, lack generalizability, have a narrow focus and scale, and have limited geographical scope. For instance, Filimonau et al. (2019b) noted that related research has largely ignored the managerial aspect of food waste mitigation. Similarly, despite the issue being more noticeable in developing economies, Papargyropoulou et al. (2019) argued that the extant literature is skewed toward developed nations. With the reduction of food waste being a crucial objective at both national and international levels due to its linkage with food sustainability (Thamagasorn and Pharino, 2019) and implications in terms of increased use of natural resources (Wunderlich and Martinez, 2018), the existence of such limited academic literature is quite concerning, and impetus needs to be given to research in the area. The embryonic state of prior studies in terms of both width and depth reinforces the exigency to bring the fragmented findings together to propel future research. We propose to address the gap through a systematic review of the literature on food waste in this sector.

Admittedly, there are some noteworthy literature reviews in the domain, but most of the relatively recent ones were focused on diverse aspects of food waste, such as households (Hebrok and Boks, 2017; Roodhuyzen et al., 2017; Schanes et al., 2018), hospitality operations (Filimonau and De Coteau, 2019), sustainability (Baig et al., 2019; Carino et al., 2020; Lemaire and Limbourg, 2019), food waste accounting (Corrado and Sala, 2018), interventions (Reynolds et al., 2019), the food supply chain (Özbük and Coşkun, 2019), solid waste management (Pirani and Arafat, 2014), and national school lunch programs 1978–2015 (Byker Shanks et al., 2017). It is evident from the above that there are limited reviews focusing specifically on food waste in the hospitality sector. Of the recent reviews, only one, conducted by Filimonau and De Coteau (2019), amalgamates the findings of studies on food waste in the hospitality sector. Their review synthesized prior findings to provide a food waste mitigation framework from the standpoint of managers in hospitality operations.

Our study uses this review as a conversant/referent article and builds on its findings in three ways: (a) by conceptualizing the hospitality sector as the hospitality and food services (HaFS) sector in light of the extant definitions and delineating it into three clear sub-domains, which can help create a more nuanced analysis, (b) by including academic research specifically on the profit sector of HaFS, and (c) by developing a framework for academic research, thereby adding a new dimension to the practice-oriented viewpoint offered by the existing review. Specifically, our main research question is how the state-of-the-art literature on food waste in the profit sector of HaFS can be used as a platform for setting the future research agenda. To address our research question, we have given equal emphasis to research profiling and content analysis of the extant literature in the area.

The synthesis from the current systematic literature review (SLR) study uncovered thematic foci and research gaps related to food waste in the profit sector of HaFS, such as the robustness of the methods used, the reliability of the samples, the geographic coverage, the demographic factors, the behavioral manifestations, the theoretical frameworks, and the narrowness of the variables explored. The extensive analysis of the selected studies resulted in the development of a framework for research and provides useful inferences for practice.

2. Hospitality and food services sector

The HaFS sector includes all outlets that serve food and/or drinks for immediate intake in an out-of-home setting (WRAP, 2013). It includes several key sub-domains, such as restaurants, hotels, health care, education, and staff catering (WRAP, 2013). Speaking broadly, restaurants include establishments serving different cuisines, such as Italian, Chinese, Indian, or French, as well as quick-service restaurants offering both eat-in or take-away at various locations, including those offering leisure. Hotels include accommodation providers such as luxury hotels, budget hotels, bed & breakfasts, and youth hostels. Health care includes hospitals, nursing homes, and care centers. Education includes preschools, primary and secondary schools, tertiary education centers, colleges, and universities. Staff catering includes canteens and cafeterias located in workplaces for feeding employees.

Many scholars have defined hospitality to include both commercial as well as social aspects (Brotherton, 1999; Brotherton and Wood, 2000). This description is congruent with recent research, where the hospitality sector has been argued to comprise commercial, non-commercial, and other food service sectors (Betz et al., 2015). Similarly, according to Marthinsen et al. (2012), hospitality comprises two sectors: profit and cost. The profit sector includes hotels, restaurants, and cafés (HORECA); canteens; and catering, such as in supermarkets, while the cost sector comprises accommodation and food service in establishments like schools, universities, health care, and other establishments where profit from hospitality services is not the key consideration. Furthermore, some prior studies have classified educational and health care establishments as institutional food waste generators (Ai and Zheng, 2019).

The hospitality sector is quite complex, as acknowledged by prior scholars (Filimonau and De Coteau, 2019), and even though there are some common tracks, nuances of food waste are different for the profit and cost sectors. For instance, while restaurants can consider food donation/charity, hospitals may not have that option due to the risk of infection. Similarly, while schools can offer left-over lunches at subsidized rates by using leftovers in serving dishes after the schoolchildren have finished (Laakso, 2017), the same may not be possible for restaurants that serve diners throughout the day. Additionally, we feel that due to the growing trend toward privately owned hospitals, elder care, retirement homes, and nursing homes, catering in such establishments may not always be a part of the cost sector. This is particularly relevant for countries where medical facilities are not government-funded. Based on this, we propose that the HaFS sector may be considered as comprising three key segments: (a) the business segment, including accommodations and food service at hotels, restaurants, cafés, workplace canteens, inflight catering, snack bars, coffee shops, and pubs; (b) the education segment, including nurseries, hobby classes, primary schools, secondary schools, tertiary education centers, colleges, and universities; and (c) the health care segment, comprising hospitals, elder care, retirement homes, and nursing homes. Based on these classifications, we set our research boundary as all food service

establishments operating under the business segment of the hospitality sector, which represents the profit sector of HaFS. Based on this, the identified keywords for the search of studies to be reviewed were food waste, food loss, food service business, waste of food, catering, restaurants, cafeteria, coffee shop, snack bar, workplace canteen, inflight catering, airlines catering, and pub.

3. Methodology

The current study adopted the SLR approach because it allows for presentation of the extant literature in the area in a comprehensive and comprehensible manner (Tranfield et al., 2003). Many recent studies have used this approach to review prior literature (Kushwah et al., 2019; Sahu et al., 2020). The SLR methodology process of review has two distinct steps, namely, data extraction and, later, research profiling (Behera et al., 2019; Seth et al., 2020). The review panel for the study, consisting of two academics with knowledge about the food sector and four researchers, was constituted to help the researchers set the conceptual boundary, arrive at a consensus after the independent coding, and to resolve issues and disagreements if any arose.

3.1. Data extraction

The data extraction stage consisted of two main components—setting the research objectives and selecting the relevant studies.

3.1.1. Setting the research questions

We divided our main research question into four parts for the purpose of analysis, and accordingly, the present SLR addressed the following four research questions (**RQs**):

RQ1. What is the research profile of the selected studies in terms of summary statistics?

RQ2. What are the different themes of the accumulated research on food waste in the profit sector of HaFS?

RQ3. What are the research gaps in the existing literature?

RQ4. How can research into food waste in the profit sector of HaFS be taken forward?

3.1.2. Selection of the studies

The study utilized two well-known databases, *Scopus* and *Web of Science* (WoS), for the selection of studies to be reviewed. Both have been used frequently in previous SLR studies (Mongeon and Paul-Hus, 2016). The keywords searching the databases were searched for included the following: ("food wast*" OR "food loss*" OR "foodservice business" OR "wast* of food") AND ("cater*" OR "restaurant*" OR "cafeteria*" OR "coffee shop*" OR "snack bar*" OR "workplace canteen*" OR "inflight cater*" OR "airline* cat*" OR "pub*").

The studies found through the search protocol were evaluated using the inclusion criteria of (a) studies on food waste generation in the profit sector of HaFS, (b) studies available on Scopus and WoS through March 28, 2020, and (c) studies published in peer-reviewed journals in English only. The exclusion criteria were (a) studies related to other areas in the domain, such as waste to energy, digestion-composting processes, anaerobic treatment and valorization of waste, and the characterization of hotel biowaste, to name a few; (b) review articles, conceptual papers, proceedings, guidelines, and discussion papers; and (c) duplicate studies.

The initial search in Scopus generated 2,008 studies. These included journal articles, conference papers, editorials, reviews, data papers, erratum, notes, short surveys, book chapters, books, and editorial letters in all languages. To begin with, 741 items other than journal articles in English were eliminated. Of the remaining 1,267 articles, 1,209 were further excluded, as they pertained to areas of food waste that are not directly connected to food waste generation in HaFS. The remaining 58 studies related directly to food waste in the profit sector of HaFS, within the conceptual boundary described in Section 2, were taken forward. The search of the *WoS* database resulted in the identification of 1,380 studies, of which 290 were items other than journal articles in English that were eliminated. Of the 1,090 studies taken forward, 525 were already listed in Scopus, so they were eliminated from the WoS list. Next, 560 studies on topics such as anaerobic treatment of food waste, valorization, and so on were eliminated. The remaining five studies were taken forward. This resulted in a common pool of 63 studies to be included in the review. The citation chaining search confirmed that no relevant studies had been missed. While selecting studies, attention was also paid to the quality of the journals that published the articles, and gray literature was excluded. The final count of studies taken forward for review was 63, which were published between 1983 and March 2020.

3.2. Research profiling

The presentation of the research profile of the selected studies includes descriptive statistics such as timelines, establishments investigated, geographic scope, research design, methods of data analysis, variables examined, and the theoretical bases. The year-wise classification of the publications suggests that fewer studies were published before 2017; after that, the research started gaining momentum, and there was a tremendous upsurge (Fig. 1). The data presented in Fig. 2 shows that the selected publications came from a wide variety of journals.

A variety of establishments, including restaurants, hotels, workplace cafeterias, and inflight catering services, were individually examined by the scholars in the selected prior studies (e.g., Charlebois et al., 2015; Sakaguchi et al., 2018, Fig. 3). However, some studies investigated a variety of food services (e.g., Heikkilä et al., 2016), and we classified those studies as mixed (Fig. 3).

The examination of the geographic scope of the prior studies suggests that the majority focused on a single country, with most conducted in Western countries and only a few coming from developing countries or multiple locations. The highest number of studies came from the United States (Fig. 4).

The current study reviewed the selected studies from the perspective of their research design as well as their methods of data analysis. Fig. 5 reveals that the researchers used mainly four types of research designs.

In addition, the prior literature used a variety of commonly applied methods for data analysis, such as regression analysis (linear, generalized linear models, logit, tobit); parametric tests (ANOVA, ANCOVA, Z-test, *t*-test); non-parametric tests (Wilcoxon ranked sum, Mann–Whitney, Kruskal–Wallis, chi-square); post hoc analysis; structural equation modeling; and thematic, content, and cluster analysis. Interestingly, only a few studies invoked any theoretical frameworks or theories to propose and test the research model. The theories applied by the selected studies were the theory of reasoned action (Hamerman et al., 2018), practice theory (Hennchen, 2019), the theory of planned behavior (Liao et al., 2018; Lorenz et al., 2017), the value–attitude–behavior (VAB) theory (Kim et al., 2019), and the norm-activation theory (Lorenz et al., 2017). Additionally, Filimonau et al. (2020a) confirmed the usefulness of the TPB- and TRA-based models in predicting pro-environmental consumer behaviour in tourism and hospitality settings.

Furthermore, the prior studies examined an interesting set of dependent variables to spotlight the key areas related to food waste generation by HaFS and the possible intervention to reduce the

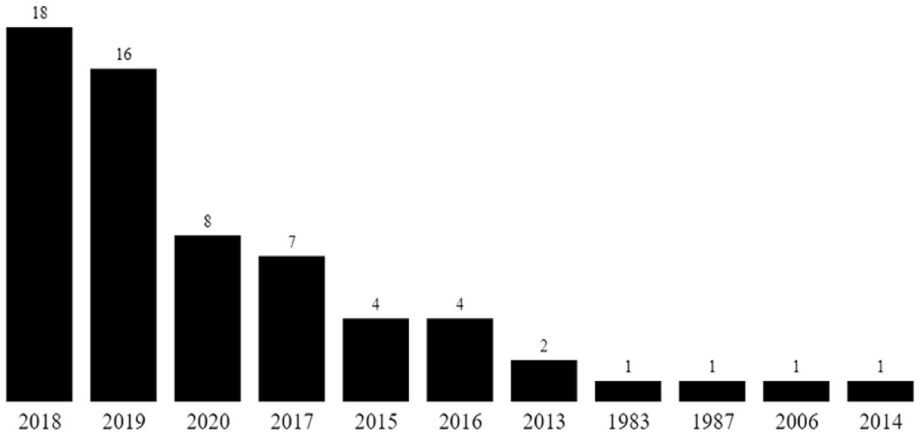


Fig. 1. Year of publication of the selected studies.

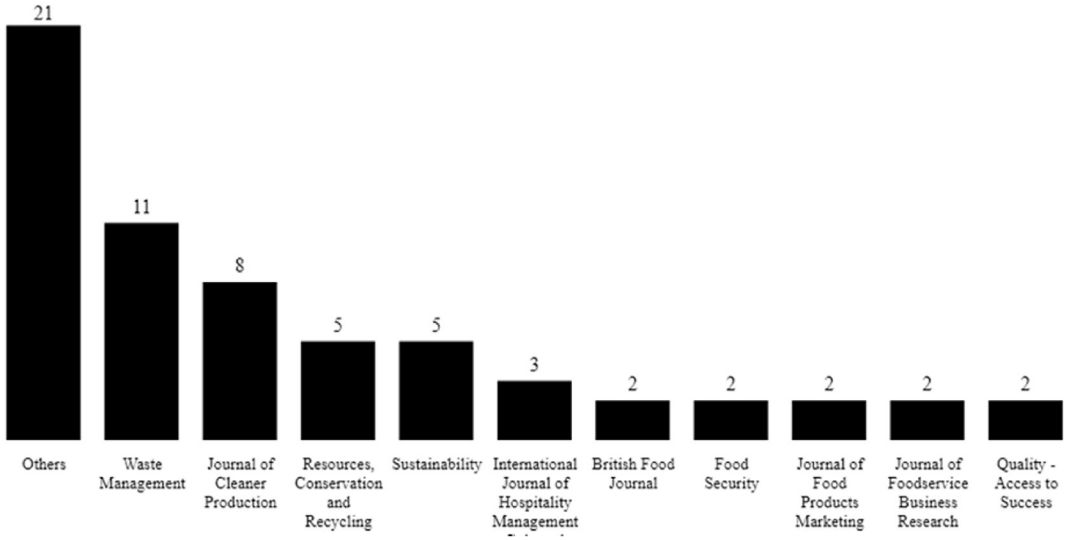


Fig. 2. Journals publishing the selected studies.
Note: Other journals that each published one study include African Journal of Hospitality, Tourism and Leisure; Detritus; Economics Letters; Environmental Research, Engineering and Management; Environmental Science and Pollution Research; International Journal of Culture, Tourism, and Hospitality Research; International Journal of Sustainable Development and Planning; Journal Of Agriculture Food Systems And Community Development; Journal of Consumer Behaviour; Journal of Experimental Psychology: Applied; Journal of Hospitality Marketing and Management; Journal of Quality Assurance in Hospitality and Tourism; Journal of Retailing and Consumer Services; Journal of Sustainable Tourism; Journal of the Air and Waste Management Association; Public Health Nutrition; Science of the Total Environment; Sustainable Production and Consumption; Tourism Management; Tourism Management Perspectives, and Waste Management & Research.



Fig. 3. Establishments examined by the selected studies.

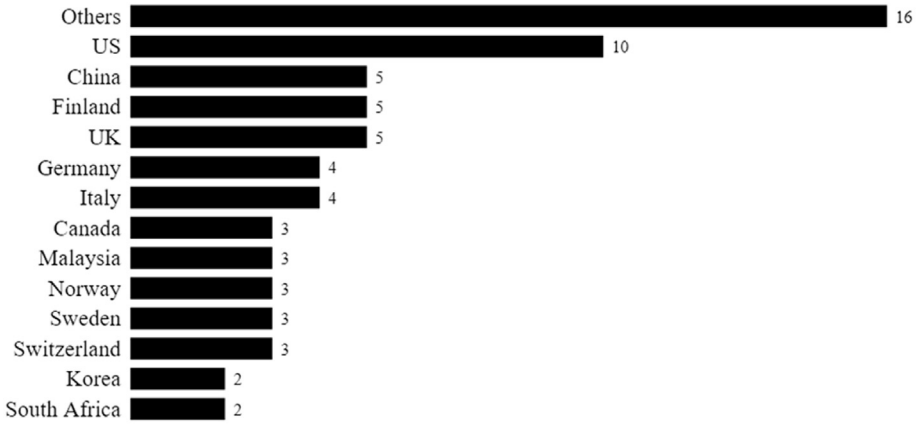


Fig. 4. The geographic scope of the selected studies (based on a sample).
Note: Others include Bulgaria, Czech Republic, Egypt, France, India, Lithuania, Malta, Netherlands, New Zealand, Pakistan, Poland, Slovenia, Spain, Thailand, and UAE (one study each). One study used data from 33 developed countries in North America, Europe, Asia, and Oceania.

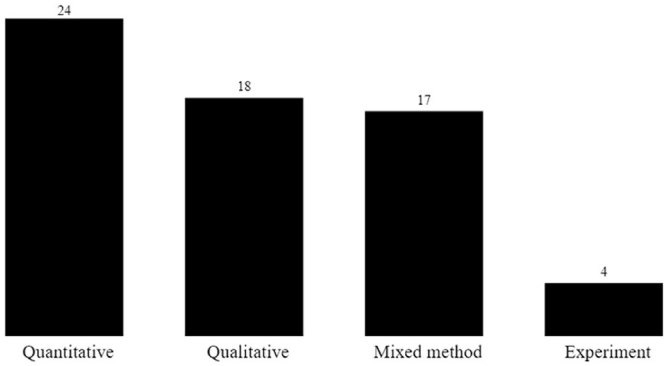


Fig. 5. The research designs used in the selected studies.

waste. Tables 1 and 2 present the key dependent variables and independent variables (categorized into four groups each), respectively.

4. Discussion

The study aimed to provide insights into the complex nature of the prior research on food waste in the profit sector of HaFS, represented by hotels, restaurants, cafés, canteens, inflight catering, pubs, and snack bars. To this end, the selected studies were analyzed and synthesized using the content analysis method, a common technique for analyzing qualitative data (Hsieh and Shannon, 2005). Two researchers independently performed the content analysis to uncover the thematic foci of the extant literature. After much debate and the seeking of opinions from two academics with knowledge of the food sector, a consensus was achieved on nine broad research themes: the causes of food waste, waste hot spots, nudges, types of facilities, the handling of leftovers and its impacts, stages of waste generation, demographic factors, and the control of food waste. The researchers had agreed on most of the themes, including hot spots, nudges, the handling of leftovers, the impact of food waste, demographic factors, and the control of food waste. However, there was some debate over the classification of the causes of food waste, waste by type of establishment, and stages of waste generation due to the overlap in their content. However, it was finally decided to present them separately to make the discussion more granular.

4.1. Causes of food waste

Understanding the causes of waste is important because effective food waste management requires a deeper appreciation of the volume and origin of waste (Ai and Zheng, 2019). In general, researchers discussed in detail the food wasted during preparation, serving, and consumption. Overproduction, serving issues, and plate waste were taken into consideration to ascertain related causes (e.g., Sebbane and Costa, 2018). The key causes were the nature of the food menu, the production procedure, and the use of pre-prepared versus whole food products (McAdams et al., 2019). Dinnerware size was also linked with food waste, with establishments using larger dinnerware and wasting more food (Wansink

Table 1
Dependent variables.

Dependent variables	
Food waste generation	Aggregate food waste
	Plate waste
	Leftovers
	Food waste production and dish use
Food waste prevention/reduction	Food waste prevention
	Food waste reduction
	Intention to prevent leftovers
Handling leftovers	Waste disposal practices
	Donation to staff and charity
	Doggy bag participation
Perspectives	Stakeholder perspectives
	Managerial perspective

Table 2
Independent variables examined by the selected studies.

Independent variables	
Food service variables: Managerial-side	Inventory management and checks
	Cold storage temperature
	Mitigation of waste
	Supplier agreement and relationship
	Management standpoint on food waste
	Business models and concept
	Resource availability
	Menu design and specifications
	Food provisioning
	Competitors
	Food sales revenue
	Environmental sensitivity
	Employee commitment and responsibility
	Accuracy in prediction of the number of expected customers
	Cooperation between hotel/restaurant staff and guests
Food service variables: Kitchen-side	Overproduction
	Size (entrees, plate, meal, portion, catering, restaurant)
	Food prepared
	Food temperature
	Meal presentation
	Cooking place
	Serving style
	Food type
	Quality assurance
Food service variables: Leftover handling	Liability concerns
	Improper disposal
	Leftover treatment
	Distribution of surplus foods
	Barriers to food donation
Food waste reduction variables: Interventions & nudges	Awareness campaigns
	Fiscal and legal instruments
	Awareness of food waste and disposal
	Financial incentive
	Pricing strategy
Consumer variables: Food waste generation	Consumer wastage
	Past experience
	Perception
	Income level
	Hunger
	Intentions towards food waste
	Attitudes and behaviour
	Social desirability and situation
	Customer expectation and satisfaction
	Demographic factors
	Eating habits
	Dining out frequency
	Purpose of the meal

and van Ittersum, 2013). Other factors identified as causes of food waste included the skill levels of employees (Kasavan et al., 2019; McAdams et al., 2019), product development and procurement (Heikkilä et al., 2016), portion sizes (Berkowitz et al., 2016; Von Massow and McAdams, 2015), culture, inventory management capacity, and sensitivity to environmental issues (Lanfranchi and Giannetto, 2017).

With specific reference to restaurants, food waste generation can be attributed to factors such as the type of ingredients used, the dishes served, and the opening hours (Principato et al., 2018). For instance, restaurants with meat-based menus that served both lunch and dinner generated more waste (Principato et al., 2018), while others reported losses due to overproduction and food spoilage (Aamir et al., 2018). On the other hand, for buffet operations, the primary causes were an inaccurate prediction of customers, a strict policy of not serving leftover food in the next meal, serving the waste, and poor coordination of various functional areas such as bookings, purchasing, and preparation with the operations department (Pirani and Arafat, 2016; Silvennoinen et al., 2019). Taste perception was a key cause of plate leftovers in workplace canteens (Lorenz et al., 2017), whereas in the case of airlines, a direct association between the flight length and food waste was observed (Blanca-Alcubilla et al., 2018).

4.2. Food waste hot spots

Scholars examined whether certain food items were wasted more than others, which are collectively referred to as food waste hot spots. The findings from the literature showed that food waste hot spots in restaurants and diners were salads and side dishes like pasta, potatoes, and rice (Silvennoinen et al., 2015). Findings also give insight into the possibility of geographic and cultural disparities in food waste hot spots. For instance, the waste hot spots in specific restaurants in Canada were starches (Charlebois et al., 2015), whereas hot spots in hotel restaurants in Malaysia were fruits, vegetables, sauces, oils, and fats (Papargyropoulou et al., 2016). In the case of inflight food service, the breakfast snack was the meal hot spot of waste, and vegetables were the food items that contributed the most to waste (El-Mobaidh et al., 2006; Thamagasorn and Pharino, 2019). Furthermore, most of the waste

was composed of the organic matter coming from menu-based items (Blanca-Alcubilla et al., 2018). All in all, there were no prominent trends in food waste hot spots to trace them back to cultural factors or geographic location, except that vegetables are likely to be a universal hot spot.

4.3. Stages of food waste generation

Food waste generated at different stages may differ in composition and may be recycled accordingly (Ho and Chu, 2018). Due to this, the diagnosis of waste generated at different stages is essential for effective food waste management. The findings suggest that waste occurred at different stages, from procurement to consumption. Scholars have adopted the practice of measuring in terms of food waste in grams per meal at the stages of storage, preparation, serving, and plate (Betz et al., 2015). Similarly, they have also classified food waste generation levels as kitchen waste (related to stages of food preparation), serving loss waste, and plate leftovers (Heikkilä et al., 2016). The findings also indicate that food waste occurring at the production stage is of two types: avoidable and unavoidable (McAdams et al., 2019). With regard to serving waste, its cause in restaurants has been shown to be related to the establishment's adherence to various quality assurance practices and standards (McAdams et al., 2019). For example, the lowest per customer food waste was generated in a limited-service restaurant with fewer standardization and quality control procedures (McAdams et al., 2019).

4.4. Food waste by the type of establishment

Scholars have delved deeply into the differences in the food waste generated by different types of HaFS establishments and concluded that it is related to the practices, conditions, or characteristics of the establishment under consideration. Some of these studies assessed the difference in food waste by considering four common restaurant types, namely, quick-service, limited-service, casual dining, and fine dining (e.g., McAdams et al., 2019; Tatàno et al., 2017). In general, casual dining restaurants have greater plate waste volume than fine dining restaurants, which have greater food waste per customer. However, “whole” products are consumed in larger quantities (McAdams et al., 2019). Along these same lines, the amount of waste per restaurant was found to be highest for the high-end restaurants, followed by medium- and low-end ones (Aamir et al., 2018). Per-employee food waste was found to be lower in larger restaurants (Tatàno et al., 2017). Another study found that the purpose of the meal and the business model also impacted the amount of waste (Katajajuuri et al., 2014). For instance, business banquets and friends’ parties in large or medium-sized restaurants led to more waste when compared to working and private meals in smaller restaurants or snack bars (Wang et al., 2017). Similarly, waste constituted approximately 30% of the food input in energy terms for hotels, whereas it was only 3% for restaurant complexes (Youngs et al., 1983). Furthermore, food wasted per portion also varied depending on the type of establishment, with approximately 50 g/portion in the case of canteens and 190 g/portion for restaurants (Malefors et al., 2019).

4.5. Demographic factors and food waste

The demographic variables of age, gender, and geographic location were found to be associated with the causes and prevention of food waste. For instance, Hamerman et al. (2018) suggested that gender differences were not significant with respect to the willingness of individuals to get leftovers packaged due to the potential embarrassment or feelings that it was counter to social norms. However, age and gender played a significant role in food waste, with females and younger diners leaving more food unconsumed (Collison and Colwill, 1987). On the other hand, Sebbane and Costa (2018) found a discrepancy between the stated and actual amounts of food wasted by individuals, with men more likely to exhibit greater differences.

On the issue of cultural and geographic differences, Katajajuuri et al. (2014) observed that the factors influencing food waste, such as food consumption patterns related to ready-to-eat versus freshly cooked food, varied according to the country, suggesting the likely impact of cultural differences. Similarly, Wang et al. (2017) compared restaurant waste in China with Western countries and found that it was nearly the same as the average waste in Nordic countries. The study also revealed that restaurant food waste was higher in the case of tourists as compared to local residents, and there were also differences in restaurant food waste by the type of city where the restaurant was located.

4.6. Impact of food waste

Food-related waste has serious implications in terms of sustainability and environmental issues. Researchers have even called food waste the unsustainability hot spot (Eriksson et al., 2018). The key concerns in this context include climate change (Kallbekken and Sælen, 2013), monetary losses (Hennchen, 2019), food security (Wang et al., 2017), and the overall economic impact of such waste (Heikkilä et al., 2016). These concerns are confirmed by estimates revealing that waste food constitutes an economic loss of 23% of food purchased (Papargyropoulou et al., 2019). Similarly, the ecological impact of such waste in HORECA in China was found to be nearly twice the size of its arable land (Wang et al., 2018). The concerns become even more confounding due to the fact that even though food service professionals were aware of the implications in terms of monetary losses and showed a willingness to reduce waste, customer satisfaction, rather than waste mitigation, was their top priority (Hennchen, 2019).

4.7. “Nudges” to reduce food waste

Nudges may be described as the signals and reminders sent to make the concerned stakeholders aware of their part in waste reduction. Nudges and other interventions have been found to be useful in decreasing food waste in cafeterias, canteens, and restaurants (Dolnicar et al., 2020; Kallbekken and Sælen, 2013; Stöckli et al., 2018). These interventions could be in the form of specific non-intrusive ‘nudges’ like reducing plate size (Tatàno et al., 2017), removing side plates (Camilleri-Fenech et al., 2020), providing social cues such as small posters to create awareness (Kallbekken and Sælen, 2013), offering to pack leftovers (Hamerman et al., 2018), and giving informational and normative prompts through place cards on tables (Stöckli et al., 2018).

The effectiveness of these nudges was different for different establishments. For instance, in restaurants, reduced plate size was more effective than social cues given through posters, and there were no adverse implications for customer satisfaction (Kallbekken and Sælen, 2013). In the case of all-you-can-eat food services, reduced plate size was also effective, resulting in a decrease of as much as 30% in plate waste (Ravandi and Jovanovic, 2019). Differently, messages related to the environment had a positive impact on consumers’ attitudes toward the reduction of such waste in buffet-style restaurants (Chen and Jai, 2018).

4.8. Handling of leftovers

Reuse and recycling of food waste are considered optimal

approaches to handling waste, with dumping and energy production being the last resort (Betz et al., 2015). Packing leftovers for taking away or the donation of unconsumed food are standard approaches for handling consumable waste (Mirosa et al., 2018; Pirani and Arafat, 2016). However, it is not a simple decision, particularly in restaurant dining. In fact, taking leftovers away as "doggy bags" is the net outcome of various barriers and benefits that impact the related consumer behavior (Mirosa et al., 2018). The obstacles related to doggy bags differ from individual to individual and arise from both conveniences as well as factors related to social stigma. For instance, shame and trying to save face act as barriers, negatively affecting the intentions to take away leftovers (Dagiliūtė and Musteikytė, 2019; Liao et al., 2018). Similarly, situational social factors also influence diners' intentions to get the leftovers packed for taking home (Hamerman et al., 2018). For example, when customers dine with someone they want to make a good impression on, they might not carry leftovers home because it might be considered a violation of social norms (Hamerman et al., 2018; Sirieix et al., 2017). At the same time, practical and moral reasons such as the fitness of leftovers for consumption and concern for the environment also impact diners' intentions to take them (Hamerman et al., 2018). Furthermore, a paradoxical behavior of consumers has been observed in this context, where on the one hand, personal norms regarding food waste cause guilt over leaving leftovers behind, and on the other hand, social norms create pressure to avoid asking for a doggy bag (Sirieix et al., 2017).

Apart from being taken by the diners, leftovers can be reused in a variety of other ways, such as distributing them among staff members and/or donating them to charity (Amato and Musella, 2017; Bharucha, 2018). However, some restaurants avoid food donations as they fear the legal liability arising out of someone complaining after consuming donated food (Sakaguchi et al., 2018). In fact, due to this fear, some hotel chains donate only untouched leftovers to charity (Pirani and Arafat, 2016). In this regard, authorities can encourage restaurants to donate food by providing certain tax incentives (Filimonau et al., 2020b). But leftovers that are unfit for consumption can still be used, and are, for composting or feeding livestock (Okumus, 2020; Otten et al., 2018). However, in airline catering, even sealed post-flight food is just disposed of along with other food waste (Sambo and Hlengwa, 2018).

4.9. Food waste control practices

There exists some level of consciousness to avoid food waste and about the use of waste mitigation practices at restaurants and other HaFS establishments. For instance, different types of restaurants are quite aware of the need to prevent food waste as it leads to financial benefits arising from savings in food costs (McAdams et al., 2019). Keeping this in mind, most restaurants try to manage waste by ensuring proper chopping and trimming of vegetables and meats, effective purchase planning, and demand forecasting (Aamir et al., 2018; Bharucha, 2018; Principato et al., 2018). Competent and skilled staff and the participatory approach of involving employees in formulating and implementing control measures are also quite useful in reducing waste (Heikkilä et al., 2016; Strotmann et al., 2017).

The role of government is also considered significant in controlling food waste. For instance, Wang et al. (2017) highlighted the active steps taken by the government in Beijing to reduce such waste, including running campaigns like "Clean Your Plate." In the context of the HORECA industry, inclusive legislative frameworks and campaigns to create awareness, along with fiscal incentives, have had a noticeable impact on waste reduction (Chalak et al., 2018). Furthermore, targeted policy interventions to increase corporate and consumer commitment are also important (Filimonau et al., 2019b) as are free training programs by the government to decrease kitchen waste (Filimonau et al., 2020c). The government's role is all the more crucial because despite knowing that a comprehensive approach to effective waste reduction requires various innovative actions, food service establishments tend to only implement innovations that are driven by cost-saving criteria (Martin-Rios et al., 2018). Furthermore, proper quantification can also serve as a benchmark for food waste prevention in various food service establishments. For instance, automated quantification tools can serve as a cost-efficient way of reducing food waste in the case of establishments reporting a very high volume of waste (Eriksson et al., 2019).

5. Research gaps and implications

Our systematic review of the prior literature provided us a bird's eye view into the varied contours and contributions of the extant findings. It also enabled us to identify several gaps in the prior findings. These gaps serve as the basis for proposing a forward-looking research agenda to motivate academic research and present actionable implications for managers and policy makers.

5.1. Research gaps

5.1.1. Data-related gaps

A total of five main gaps were identified, including problems related to generalizability, research design, self-reports and voluntary surveys, sample sizes and representativeness, and data collection issues.

a) **Generalizability problems:** The selected studies covered a wide variety of geographic locations, food service establishments, and contexts. This has enriched the literature in a specific manner, but at the same time, such a diversity of contexts has also raised issues about the generalizability of study findings to other contexts (e.g., Betz et al., 2015; Stöckli et al., 2018).
b) **Research design issues:** Many of the selected studies utilized a qualitative research design to examine food waste (e.g., Bharucha, 2018). The qualitative design suffers from inherent problems that can adversely affect research outcomes. These problems include the formulation of research questions, ethical considerations, contextual issues, and non-numerical data analysis leading to questions about the rigor of the results (e.g., Beaulieu and Estalella, 2012; Gelling, 2011; Nye et al., 2016). Similarly, some studies employed experiment-based designs that were conducted in hypothetical scenarios rather than in real-world settings, thereby imposing a limitation on the findings (Hamerman et al., 2018).
c) **Self-reports, voluntary surveys, and interviews:** The use of semi-structured interviews (e.g., McAdams et al., 2019) and cross-sectional studies (e.g., Sebbane and Costa, 2018) were common in the literature. These methods have inherent weaknesses, such as participants in self-reported surveys remaining guarded in their responses and not always giving factual replies (e.g., Bharucha, 2018; Liao et al., 2018). Similarly, the problem of selection bias could lead to bias in collected responses (e.g., Hamerman et al., 2018; McAdams et al., 2019).
d) **Data collection issues:** Many of the selected studies relied on small sample sizes to collect data (Betz et al., 2015; Chen and Jai, 2018; Stöckli et al., 2018). The use of smaller sample sizes in relation to the size of the target population raises issues related to the representativeness of the sample and, thus, the reliability of the findings.

5.1.2. Gaps related to analysis

A total of five main gaps were identified, and these include the narrow focus of prior studies, underestimation of the quantity of waste, limited geographic scope, limited breadth of analysis in terms of mediating and moderating variables, and lack of theoretical frameworks.

a) **The narrow focus of prior studies**: The examined studies took a narrow focus on the issues related to food waste (e.g., Katajajuuri et al., 2014; Sebbane and Costa, 2018). For instance, Kallbekken and Sælen (2013) discussed nudges for reducing waste and their implementation, but they did not test the implementation of the two proposed nudges—information and reduced plate size—together. Testing the impact of the simultaneous implementation of both these nudges could have improved the understanding of the practitioners since they would prefer to address food waste through a multipronged strategy. Similarly, many studies discussed prompting food saving behavior, but the underlying psychology, as well as the long-term effects of informative and normative prompts, were not well-examined (e.g., Stöckli et al., 2018). Moreover, consideration was given to only a small number of variables relating to food waste and waste hot spots, and as a result, the real-life validity and usefulness of the findings are severely restricted.

b) **Underestimation of the quantity of waste:** There are varied issues related to how the generated waste is quantified. This is a serious issue, as the lack of proper quantification can lead to an incorrect assessment of waste (Wang et al., 2017). For instance, weighing plate waste at the individual level in busy dining establishments can be quite misleading. Since accurate quantification of waste is a key aspect for identifying waste hot spots and initiating waste prevention measures, the issue of determining the right quantity of waste represents a key concern for research in the area.

c) **Limited geographic scope**: The reviewed literature largely focused on developed or industrialized countries, as presented in Fig. 4, with very few studies conducted in emerging countries, as argued by Wang et al. (2018). This represents a gap in the research because food waste is a grave challenge in developing countries that are battling with rising population numbers, urbanization, and low incomes. This finding reflects the need for more food waste studies in developing countries.

d) **The limited breadth of analysis in terms of mediating and moderating variables**: Although it has been contended that food waste from out-of-home dining is dependent on personal characteristics, social settings, and situational variables (Betz et al., 2015), the majority of the prior studies have not examined the mediating or moderating influences of such factors. For instance, the type and time of a meal can potentially mediate the association between the type of restaurant and food waste generation, but it has not been investigated. Limited studies have considered factors that may have a moderating influence on waste reduction behavior. For instance, Kim and Hall (2019) confirmed the moderating influence of dining expenses on the association of attitude with intentions to reduce waste. But there is a noticeable gap in terms of considering the moderating effects of demographic variables such as age, gender, and country despite acknowledgment of their influence on food waste generation or prevention behavior (e.g., Hamerman et al., 2018; Silvennoinen et al., 2015).

e) **Lack of theoretical frameworks:** Most of the selected studies were largely based on the qualitative and survey-based approach to collecting data from food service establishments. Although the findings were interesting and useful, most of the studies did not utilize any theory or theoretical framework. Thus, they lacked the theoretical underpinnings that would make them much more useful for guiding further studies. Only a few of the studies from the selected pool utilized some kind of theory. Given the fact that food waste generation and prevention are the manifestations of consumer behavior, this area can benefit from the use of different behavioral theories and frameworks.

5.2. Potential research areas

The research gaps identified by us serve as the basis for recommending potential areas where future researchers can contribute to theory and practice, as discussed in upcoming sections. The key suggestions are related to testing different geographies and food service establishments, advances in quantification methods to measure food waste, utilization of diverse research methods and variables, deeper diagnoses of interventions and nudges to create awareness about food waste, improved theoretical perspectives, and the inclusion of online food delivery (OFD) platforms in food waste research.

5.2.1. Testing different geographies and food service establishments

The generalizability of findings of the prior studies was a major concern. To address this gap, scholars can undertake comparative and replication studies by testing the existing research models in different contexts. For example, a model developed and tested in the context of restaurants can also be examined for cafeterias and workplace canteens. Similarly, the model tested in the context of a developed country can be suitably modified and tested in a developing economy. These comparative and replication studies can also test the efficacy of efforts to reduce waste in food service establishments. For instance, the model proposed by Sebbane and Costa (2018) can be replicated, compared, and tested in different countries to capture the impact of different cultural and policy-related factors on food waste.

5.2.2. Advances in quantification methods to measure food waste

We recommend that researchers should focus on aggregate waste data collection and utilize more comprehensive methods of food waste quantification. Thus, methods like material flow analysis (MFA), which measures the percentage of food waste generated at different levels of the catering process (Pirani and Arafat, 2016), may be used. Furthermore, policy efforts should also be directed at determining benchmarks for waste quantification so that there can be standardization in capturing the food waste data. This would also make it easier for future researchers to conduct comparative studies across various food service establishments. Extended literature has also underscored the need for better quantification of food waste as the first step toward waste reduction (Corrado and Sala, 2018). Scholars can produce more robust results and can develop more effective food waste reduction strategies by capturing food waste at various stages, such as growing, processing, production, and consumption, as suggested by Betz et al. (2015).

5.2.3. Utilization of diverse research methods and variables

We recommend that scholars should use data sets on food waste, where available, through government or other authorized data sources for analysis rather than collecting food waste data from randomly selected respondents with self-reporting instruments. In this way, the data would not be collected only with respect to kitchen waste or only from managers but also from the kitchen staff, as discussed by Principato et al. (2018) and

Silvennoinen et al. (2015). Self-reported surveys should be largely used to investigate and understand the behavioral aspects of diners and staff at food service establishments (Lorenz and Langen, 2018).

Furthermore, scholars should approach food waste from a multidisciplinary point of view and conduct mixed-method research to present a complete picture of food waste in the area. For example, investigations can focus on the causes, composition, nutritional and attitudinal aspects, ecological implications, economic consequences, and the efficacy of interventions in reducing food waste in different establishments. This is in consonance with the prior findings, which revealed that food waste quantity, as well as its composition, varies with the type of business model in the case of restaurants (e.g., Tatàno et al., 2017; Wang et al., 2017). Moreover, newer variables such as food choices, health-related food preferences, food safety concerns, hygiene consciousness, and various factors directly related to food service (Lorenz et al., 2017), as well as measures of satiety (Sebbane and Costa, 2018), should also be examined in detail.

5.2.4. Deeper diagnoses of interventions and nudges to create awareness about food waste

We recommend that scholars should focus on the development and testing of different interventions and nudges to reduce waste. In consonance with prior extended literature (e.g., Mirosa et al., 2016), we suggest that the following strategies and nudges should be evaluated by future researchers: pre-portioning dishes, removing fixed-plate choices and self-service, keeping dishes as testers for sampling before ordering, keeping self-pack boxes on each table to encourage doggy bags, removing the fixed entry fee, offering pay by weight pricing of food, and reducing the number of food choices.

Scholars should also test the efficacy of interventions such as awareness campaigns, normative nudges, guerrilla actions, and possible governmental initiatives being used simultaneously. Such investigations can provide rich insights for developing a multi-pronged strategy to mitigate waste in HaFS. Furthermore, future studies should compare interventions to highlight the effectiveness of certain nudges over others. Additionally, researchers should evaluate the costs and benefits of various food waste reduction nudges to provide useful inputs to food service managers. More information on the cost-effectiveness of interventions to reduce food waste and loss could be very useful (Read et al., 2020).

5.2.5. Improved theoretical perspectives

To overcome the paucity of theory-driven research in the domain, we recommend that future studies should utilize newer theories to study food waste related behavior and issues. For example, theories like social identity theory (Tajfel and Turner, 2004) and social–cognitive theory (Bandura, 1986) could be utilized for these purposes. Although some studies have confirmed the association between food waste reduction intentions and attitudes of consumers (Kim and Hall, 2019), there continues to be a dearth of behavioral studies in the area, particularly those dealing with attitudes and beliefs related to the use of doggy bags. Furthermore, future scholars can also contemplate extending the well-researched consumer behavior theories to explain food waste by consumers as well as food services establishments. For instance, the innovation resistance theory (Ram and Sheth, 1989) can be applied to examine the resistance of consumers to take away leftovers in doggy bags.

5.2.6. Inclusion of online food delivery (OFD) platforms in food waste research

Opening a new and more contemporary avenue for research, we recommend that the study of food waste behavior in restaurants be extended to cover OFD platforms, including food delivery apps (FDAs) that provide the facility of ordering food for delivery from a variety of food service establishments. Understanding the behavior of consumers in over-ordering the much-discounted food from these apps would be quite interesting.

5.3. Theme-based research questions

In addition to the recommendations to address the visible gaps in the extant literature, the present SLR identified certain research questions related to the nine themes that future researchers should contemplate investigating to enrich the literature in this area (Table 3).

5.4. Implications for practice

Our study offers seven implications for practice based on a critical review of the literature and the identified gaps. First, we recommend that policymakers should build a central body/data bank that is responsible for the logging and management of food waste data. It should be made mandatory for all food service establishments to enter their food waste data at a specified interval. Such a data bank could help policy makers create a ready resource for firms striving to mitigate food waste at their establishments. To make such an initiative viable and to ensure privacy and security, the data could be made available at a cost after due encryption and screening of names.

Second, a chain of food banks can be established by the government and auctioned to nongovernmental organizations (NGOs) for management at the city level. All food service establishments should be required to mandatorily take membership in food banks by paying a fee, which could be made tax deductible. This would enable businesses to save the costs they would have incurred in managing leftovers. Clear processes should be laid down to protect each party from any arising liability. This should help overcome the resistance shown by food service establishments to donating leftovers, as discussed by Aamir et al. (2018) and Pirani and Arafat (2016).

Third, to overcome the challenges associated with donating edible leftovers to charity (e.g., Hamerman et al., 2018), governments can formalize the food donation process through NGOs by involving them in running leftover meal services. Furthermore, NGOs can be channelized to act as the collecting and processing body for the disposal of inconsumable leftovers and other wastes through composting and landfill. This could be particularly useful in the case of small food service establishments that do not recycle food waste due to cost, the limited quantity of waste, and less space (Michalec et al., 2018). The government can also involve NGOs in awareness campaigns to reduce food waste.

Fourth, food service establishments can introduce a “heroes of the month” concept where the pictures of all diners who leave no leftovers on their plate are displayed, and incentives, like free dinner with the family, can be offered to one hero through a lucky draw every two months or so. Similarly, a “saver of the day” badge could be bestowed on the staff member who makes a special effort to reduce food waste by either engaging with the diners to pack leftovers or by being skilled in the handling service or in preparing the meal. These suggestions are consistent with the argument found in the literature demonstrating that the involvement of kitchen and serving staff can contribute to reducing food waste generation (Betz et al., 2015; Strotmann et al., 2017).

Fifth, the literature also suggests that buffet meals generate high food waste. Thus, food service establishments should try to promote à la carte service (e.g., Pirani and Arafat, 2016) by offering certain consumer freebies, such as free dessert when a main course

Table 3
Setting a future research agenda.

Thematic Foci	Description	Potential Research Questions (RQs)
Causes of food waste	Types of kitchen and consumer food waste	In what way can the knowledge of the causes of food waste help to reduce food waste? Does the kitchen food waste vary with the type of meal, the physical layout, the weather, and mealtime? Does understanding the causes of food waste help to better illustrate the behavioral manifestations of kitchen staff and diners?
Food waste hot spots	Food types that are generally among leftovers	Is it worthwhile to expend effort in undertaking composition analysis of the waste collected? Can the knowledge of food waste hot spots aid in developing a better menu and portion size? Is there a cultural aspect that influences the kind of food waste hot spots in different countries or the hot spots that are influenced by the type of food service establishment?
Stages of food waste generation	Waste at different levels, from procurement to consumption	Does waste at one level of the process have a multiplier effect on the waste at the other levels? Given the expense of cold storage facilities, what are the issues in using local produce delivered fresh every day? What would be the outcome of offering á la carte service instead of buffet-style meals in hotels and restaurants in terms of lost business, loss due to food waste, and inaccurate prediction of the number of diners?
Food waste by the type of establishment	The type and amount of waste generated is related to the business model, practices, conditions, and the characteristics of different types of food service establishments	Should generic models be developed for demand forecasting based on the type of establishment (e.g., casual restaurant, fine dining restaurant, and so on)? Is waste per customer and employee also related to the type of establishment, or is the variation observable only in aggregate waste? Is the variation in food waste in different types of restaurants related to menu choices, the profile of the diners, and the level of training of the kitchen staff?
Demographic factors and food waste	Linkage of food waste quantity and composition with demographic factors	Does the composition of food waste change with age and gender? Should different menus be developed based on demographic factors to reduce food waste? What are the linkages between culture and the propensity to waste food? Can Hofstede's cultural dimensions be explored in connection with food waste behavior?
Impact of food waste	Implications of food waste in terms of sustainability and environmental issues	What is the level of awareness among consumers about the link between food waste and nutritional loss? Are staff and consumers aware of the impact of food loss on food security?
"Nudges" to reduce food waste	Interventions introduced by food service establishments and government to reduce food waste	Can the same nudges be introduced at all HaFS establishments to reduce food waste? What is the efficacy of awareness campaigns in reducing food waste at different food service establishments? Is the role of government intervention in the form of fiscal and legal measures desirable from the perspective of food service establishments and consumers?
Handling of leftovers	Utilizing edible food waste through doggy bags and donations.	Can the behavior of the consumer in the context of doggy bags be better explained through behavioral theories other than the theory of planned behavior? Do financial incentives and freebies actually work to promote the acceptance of doggy bags? What has been the experience of NGOs running food banks for donating edible leftovers to charity? What are the issues in donating food to charity? Is the leftover meal service a viable option for food service establishments to utilize edible leftovers?
Food waste control practices	Food waste prevention and waste management methods	Do consumers respond negatively to the reduction in plate size in restaurants as a food waste prevention method? Would making composting mandatory lead to higher rates of disposal of edible leftovers due to convenience?

is ordered or discounts on future meals (e.g., Bharucha, 2018). The savings from the amount of food left unconsumed should compensate for the freebies. Moreover, given that freebies in the form of food items may lead to concerns, such as about obesity or health issues, gifts could also be in the form of a gym membership or some other health program, where feasible.

Sixth, all food service establishments should be made responsible for training their kitchen and serving staff in different food waste reduction methods and approaches. At the time of joining, all staff members should be required to have formal certification in waste reduction methods from a government authority. A financial penalty should be imposed on establishments failing to check the certification of their staff at the time of their employment. This is in keeping with the emphasis that earlier scholars put on the importance of the regulatory role that government can play in food waste reduction (Chalak et al., 2018; Filimonau et al., 2020c) and the importance of training employees (Gössling et al., 2011; Marx-Pienaar et al., 2020).

Finally, the gaps identified in the literature also serve as the basis for managerial implications, as summarized in Table 4.

Table 4
Research gaps and managerial implications.

Research gaps	Managerial Implications
Data-related issues (e.g., sample size, representativeness, and research design)	The available research findings are not robust and generalizable, so practitioners need to substantiate the implications with more inputs
Underestimation of the quantity of food waste	Food service businesses will benefit from having standardized measures for waste quantification and reporting at the employee, customer, and aggregate levels
Lack of comprehensive and simultaneous testing of various nudges and interventions to reduce food waste at both the staff and consumer levels	Knowing the kind of nudges and interventions that are effective in tandem for their type of establishment can help managers increase the efficacy of decisions related to the choice of nudges and interventions
A limited number of behavioral variables and food waste hot spots have been explored	Knowledge about satiety, food choices, health-related food preferences, personal factors, and the variety of food waste hot spots can help managers in developing menus more effectively and determining portion sizes to reduce food waste
Less focus on demographic variables (e.g., age, gender, income, and culture)	The information related to demographic variables can aid managers in determining portion size, the kind of nudges to be used, and waste management in terms of food donations, landfills, and composting

6. Framework development

The present study utilizes the insights obtained from the SLR to propose a framework that provides a multidimensional view on food waste and food loss in the profit sector of HaFS (Fig. 6). The objective of the framework is to provide an overview of all the stakeholders related to waste generation/reduction at any food service establishment representing the ecosystem of HaFS food waste. The framework is named the "food waste and food loss in the HaFS" model (FWFLH model). The FWFLH provides a 360-degree view of food waste in the profit sector of HaFS, spanning procurement of ingredients to disposal of edible and inedible food waste. Five stakeholders, namely, managers, staff, diners (consumers), government, and NGOs, form the core of FWFLH.

The key stakeholders in the framework are managers, staff, and diners. In our framework, these three stakeholders are posited to be the beginning point of both food waste generation and reduction efforts. Managers are involved at the procurement level, the staff at the preparation and serving levels, and the diners at the consumption level of the food supply chain. We propose that each of these three stakeholders contributes to food waste generation and reduction through (a) functional decisions, and (b) behavioral manifestations. Further, the variables associated with these three stakeholders are the independent variables (antecedents) in our framework and represent the point of data collection. An indicative set of variables and research design that can be used by future researchers to measure the food waste related behavior of managers, staff, and diners is presented in Table 5. It is important to note here that future researchers can utilize

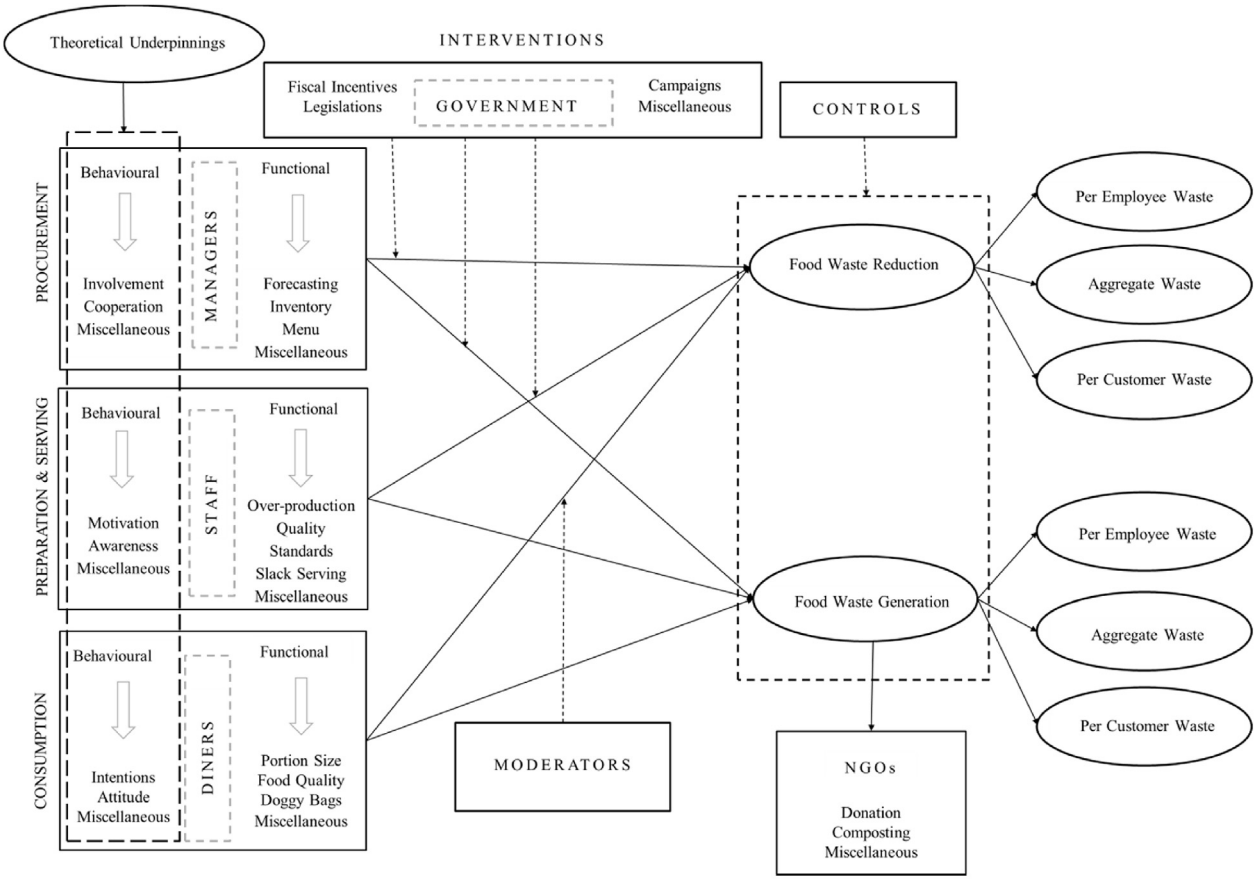


Fig. 6. Framework for food waste and food loss in HaFS (FWFLH model).

the entire framework or its parts for their studies, depending on the research objectives they are pursuing.

In addition to the variables presented in Table 5, the influence of moderating and control variables should also be considered by future researchers. In this context, researchers should collect various demographic details, such as age, gender, income, and education level of all three types of respondents (managers, staff, and diners). These variables can be utilized as moderators to measure the effects of individual differences in varied functional and behavioral factors in association with food waste generation/reduction. The influence of demographic factors is important, as revealed by prior research. For instance, it has been revealed that variations in plate waste are due to differences in the calorie requirements and appetites of male and female diners (Betz et al., 2015; Hamerman et al., 2018).

Furthermore, we suggest that future research test the framework by controlling for variations coming from other factors such as the type of restaurant (fine dining, casual dining, quick-service, or limited-service) and meal (breakfast, lunch, and so on). Differences in food waste generation based on the type of restaurant, its location, meal served, and the type of meal have been noted by prior researchers (e.g., McAdams et al., 2019; Tatàno et al., 2017). Testing the framework with and without these controls may prove to be insightful.

Table 5
Food waste generation and reduction.

Stakeholder	Manifestation	Variables	Research Design	Theoretical Underpinnings	Authors
Managers	Functional	Frequency and methods of forecasting ingredients required for cooking	In-depth interviews Questionnaire survey		Principato et al. (2018); Filimonau et al. (2019a); Filimonau et al. (2020b)
		Purchase planning, storage and supplier agreements			Bharucha (2018); Charlebois et al. (2015)
		Menu design and specifications			Principato et al. (2018)
		Method of estimation of the number of expected customers			Pirani and Arafat (2016)
	Behavioral	Increasing staff commitment and involvement	Open-ended essay	Hersey-Blanchard Situational Leadership theory (Hersey and Blanchard, 2013) Transactional contingent reward leadership (Podsakoff et al., 1982) Transformational leadership (Bass, 1985).	Strotmann et al. (2017)
		Ensuring cooperation between staff and the guests			Pirani and Arafat (2016)
Staff	Functional (Kitchen)	Frequency of over-production	Semi-structured interviews with staff In-depth interviews with managers		Aamir et al. (2018)
		Mishandling of food items	Observation		Bharucha (2018)
		Discarding dish due to bad taste, errors in preparation and/or lack of balance in ingredients			Okumus (2020); Papargyropoulou et al. (2016)
		Spoilage due to bulk purchase of ingredients or lack of proper cold storage facility			Aamir et al. (2018); Bharucha (2018); Principato et al. (2018)
	Functional (Serving)	Spillage of a dish/drink due to slack serving	Close-ended questionnaire survey with diners on multiple days, choosing different mealtimes Observation		Silvennoinen et al. (2015); Betz et al. (2015)
		Inconsistency in portion size/sloppy meal presentation			
	Behavioral	Dissatisfaction	In-depth interviews with self, peers and supervisors	Herzberg two-factor theory (Herzberg, 1971) Skinner's reinforcement theory (Skinner, 1974) Vroom's expectancy theory (Vroom, 1964)	Strotmann et al. (2017); Marx-Pienaar et al. (2020)
		Low motivation			
		Carelessness			
		Lack of awareness about food waste			
		Lack of commitment			
Diners	Functional	Large portion size	Close-ended questionnaire survey with diners on multiple days, choosing different mealtimes Observation		Betz et al., 2015; Charlebois et al. (2015)
		Poor quality of food			Lorenz et al. (2017)
		Unwillingness to take leftovers home as doggy bags due to social norms			Hamerman et al. (2018)
		Unwillingness to take leftovers home as doggy bags due to the lack of suitability of the leftover for future consumption			Liao et al. (2018); Mirosa et al. (2018); Sirieix et al. (2017).
	Behavioral	Intentions towards food waste	Cross-sectional survey	Innovation resistance theory (Ram and Sheth, 1989) Theory of planned behavior (Ajzen, 1985) Behavioral reasoning theory (Westaby, 2005)	Chen and Jai (2018)
		Attitude towards food waste			Lorenz et al. (2017); Sirieix et al. (2017)

The actions of managers, staff, and diners have an impact on the dependent variables in our framework: (a) food waste generation and (b) food waste reduction. Food waste is generated at all three levels, from procurement to consumption. It can be measured as the aggregate of the total waste generated. The prior literature has considered the aggregate waste generated as one of the important dependent variables (McAdams et al., 2019). Some studies have also measured per customer as well as per employee waste (e.g., Tatàno et al., 2017). Researchers utilizing this framework may collect and weigh waste for any one of the measures, meaning, for example, per customer or employee. Where such intensive collection is not possible, buckets could be placed for aggregate waste collection from three sources: kitchen, serving, and plate. With regard to waste reduction, some scholars have developed various interventions and nudges that can motivate managers, staff, and diners to consciously reduce food waste (e.g., Hamerman et al., 2018; Heikkilä et al., 2016; Liao et al., 2018). Researchers can conduct a longitudinal study by introducing some combination of interventions such as posters and reduced plate size and measure the pre- and post-intervention waste generated to determine the efficacy of multiple interventions introduced together.

The other two stakeholders that form a part of the framework are government and NGOs. Governments can intervene to reduce the waste generated at various levels, while NGOs can play a role in leftover reuse and disposal, such as with donation or composting. The roles of these two stakeholders can also be investigated by future researchers. In the case of government-related interventions, researchers can include questions in staff interviews as well as diner surveys to collect their views on the impact of government-run public awareness campaigns (Chalak et al., 2018; Wang et al., 2017) and fiscal incentives such as tax savings for food donations (Chalak et al., 2018; Filimonau et al., 2019b). Additionally, researchers can also examine the efficacy of other government interventions such as the imposition of a penalty on food service establishments for failing to implement proper food waste prevention and management processes, compulsory use of the specified government portal for the logging of food waste data, and the introduction of mandatory food waste certification for managers and staff. In the context of the role of NGOs in the disposal of food waste, exploratory studies on leftover meal services, food banks, and the allotment of composting contracts can be conducted.

7. Conclusion

The present study adopted the SLR methodology to distill, sift, review, and integrate extant research on food waste in the profit sector of HaFS. It curated the findings and research context of the previous literature to provide a platform for advancing further studies in the area, with specific reference to the behavioral aspects that remain least understood, as revealed by Sakaguchi et al. (2018). The SLR also offered several useful research- and practice-related inferences.

The results presented in this article are based on an SLR conducted on March 28, 2020, of published studies that met this study's scope and excluding studies that did not fit that scope. The study utilized a standard protocol and key digital databases (*Scopus* and *WoS*) to search for and identify the relevant studies. A rigorous selection process was ensured by explicating the conceptual boundary of the study clearly and using a large set of keywords for the literature search. The execution of the search protocol resulted in identifying a total of 63 studies that used various research designs, including mixed-method, qualitative, quantitative, and experiment-based.

We addressed our main research question related to setting the future research agenda in the area by dividing it into four parts (RQ1 through 4). To present an overview of the status of research on food waste in the profit sector of HaFS and address **RQ1**, the research profile of the selected studies was presented through the descriptive statistics on various aspects of the pool of publications. This included the presentation of the yearly trend of the publications, the journals publishing research in the food service field that were examined, the geographic scope of the studies, the research designs used, methods of data analysis, independent and dependent variables, and theories employed. A content analysis of the findings from the previous literature resulted in the identification of nine distinct research themes, which addressed **RQ2**. Critically evaluating the research profile and thematic foci of the selected studies answered **RQ3** by uncovering the research gaps in the extant knowledge. **RQ4** was addressed by presenting different actionable recommendations for research and proposing a framework to investigate the multidimensional nature of food waste in the profit sector of HaFS.

7.1. Limitations of the study

The current study suffers from three limitations: First, the search protocol and the different inclusion and exclusion criteria were designed based on an extensive understanding of the area and subject. However, it is possible that the keywords used might not have been exhaustive, leading to the possibility of some relevant studies being left out. Second, the selection of the relevant studies was confined only to peer-reviewed journal articles published in English. This could have led to the exclusion of relevant studies published in other languages and from other sources (such as conferences, reports, and reviews). Third, the scope of the present study was limited to the profit sector of HaFS that deals with out-of-home dining, and thus, the current study ignored a fast-growing segment that orders food from restaurants for consumption at home through the use of delivery apps.

CRediT authorship contribution statement

Amandeep Dhir: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Shalini Talwar:** Methodology, Formal analysis, Investigation, Writing - original draft. **Puneet Kaur:** Methodology, Formal analysis, Investigation, Supervision, Writing - review & editing. **Areej Malibari:** Validation, Writing - review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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